
Retention You Can Predict, Scope and Delete: a (μ, γ, T) Law and a Structural Deletion Guarantee for a Physics-Structured Memory Store

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Abstract

Existing memory systems enforce retention via explicit bookkeeping policies, such as decay factors, expiry timestamps, or learned forget gates, which dictate when a specific entry expires. However, these programmatic rules do not specify the physical rate of value degradation, the continuous dynamical effect of tuning retention parameters, or the exact residual state of the network following an item’s removal. In this work, we resolve these ambiguities by formalizing forgetting as a dynamical property of a physics-structured memory store—specifically, a damped symplectic step of a learned Hamiltonian. Our retention policy is defined by a computable physical budget parameterized by three variables: the stored direction’s spectral mass μ , the damping γ , and the temperature T .

We demonstrate three core properties of this architecture. First, the retention dial possesses a computable optimum: retention half-life is non-monotone in friction, forming a V-curve minimized at $\gamma_{\text{crit}} = 2\varepsilon\mu$. This curve spans approximately 11 orders of magnitude in μ^2 and holds across both designed and emergent (learned-MLP) units. Second, retention is rigorously scopeable: at $T > 0$, friction preserves memory while temperature erases it. A localized friction hole thus acts as a memory vault, demonstrating a $107.77 \pm 4.78\times$ retention factor on designed architectures. Finally, removal is purely structural: under canonical Blecloch & Golovin (2007) placement, store-level deletion is exact. Post-deletion, the store is byte-identical to a state that never held the item, governed by explicitly defined operational conditions, on a designed, non-learned 3-dimensional datastore at capacities 8–64.

1 Introduction

To contextualize the operational constraints of modern memory systems, we begin with deployed retention policies. Every deployed memory store currently operates under a distinct retention mechanism: a decay factor, an invalidation timestamp, a learned forget gate, or an explicit DELETE call. Despite these mechanisms, forgetting remains a critical failure mode that standard benchmarks often fail to adequately measure (Yang, 2026; Uddin et al., 2026). A programmatic policy is not a fundamental physical law. For instance, one cannot intrinsically read the half-life of a stored value off a trained LSTM, identify the exact mechanism that tunes this half-life, or verify the latent mathematical residue once a value is removed.

This gap in fundamental understanding is widest at the point of data removal. Contemporary agent-memory research supplies robust mechanisms for writing, consolidating, and retrieving data. However, for deletion, the standard approach relies on superficial bookkeeping—such as a timestamp or a dropped row (Rasmussen et al., 2025; Chhikara et al., 2025)—which leaves measurable residue within the network architecture after the fact (Chakrabortii et al., 2026; Wang & Zhang, 2026). As

the 2026 evaluation wave increasingly demands rigorous update-and-deletion tests (Yang, 2026; Uddin et al., 2026), these benchmarks require fundamentally sound mechanisms to exercise.

Therefore, in this work, we propose and analyze a memory framework where forgetting is an intrinsic dynamical property of the store itself, rather than an external bookkeeping rule. By modeling the store as a physical system, we provide computable answers to three core operational questions:

1. What is the exact retention policy, and where does its optimum lie? (Sec. 3.1)
2. Can retention be specifically scoped to a single localized item? (Sec. 3.2)
3. Is structural deletion truly absolute, and what information mathematically leaks? (Sec. 3.3)

Methodological Foundations. The memory store is constructed as a physical system. The state space (q, p) advances via a damped symplectic step of a learnable Hamiltonian, defined as $H = T(p) + V_\theta(q)$. The damping mechanism, defined as $p \mapsto (1 - \gamma)p$, inherently supplies the forgetting function, while Langevin noise introduces the temperature T . Memory items can optionally be written as *atoms*—localized energy wells superposed into a singular energy function that is subsequently read by dynamical relaxation. Our reference architecture is the Causal Learning Unit (CLU), introduced as CHLU by Jawahar & Pierini (2026). The core of this unit is exactly solvable: when diagonalized within the mass-whitened Hessian, a single damped step acts on each normal mode as a 2×2 matrix entirely fixed by $(\varepsilon\mu, \gamma)$. The spectral radius of this matrix dictates that specific mode’s retention capacity (Anonymous, 2026).

Scope and Assumptions. To clarify the boundaries of our empirical claims, we define the following operational assumptions and constraints:

- **Scale:** All evidence relies on a precisely designed store measured at a laptop-scale compute budget. This is not a deployed, large-scale LLM agent memory, nor a generalized system benchmark.
- **Claim Classification:** We report a fundamental mechanism bounded by measured physical laws, not an end-to-end system performance result.
- **Nomenclature:**
 - *Inertial mass* (M_i): The learned diagonal of the kinetic term, where larger values result in slower dynamics.
 - *Spectral mass* (μ_k^2): Defined as $\lambda_k(M^{-1/2}\nabla^2 V_\theta M^{-1/2})$, where larger values indicate shorter memory retention. Retention claims always utilize μ .
 - *Stored direction*: A written value occupies a near-flat coset direction.
 - *Integration step* (ε): Fixed at 0.05. It is never utilized as a tilt.
- **Evaluation Metrics:** Half-lives ($n_{1/2}$) are operationally meaningless without defining a specific read tolerance (Δ) and the thermal persistence length (ℓ_θ). Therefore, no half-life is quoted without its corresponding Δ and the ratio ℓ_θ/Δ .
- **Thermodynamic Consistency:** Every claim regarding $T > 0$ strictly requires the fluctuation–dissipation-consistent noise scale $\sigma_i^* = \sqrt{M_i T \gamma (2 - \gamma)}$ utilized alongside a Newtonian kinetic mode.

2 Related Work

Retention Policies in Deployed Memory Systems. Agent memories operate as external data stores that a model actively writes to, retrieves from, and dynamically edits (e.g., MemGPT, Packer et al., 2023; Mem0, Chhikara et al., 2025). Alternatively, bounded compressive states exist internal to the network (e.g., Infini-attention, Munkhdalai et al., 2024). The retention policies governing these systems rely on scores and gates. Examples include a 0.995 recency decay factor in ranking heuristics (Park et al., 2023), Ebbinghaus decay functions applied to memory strength (Zhong et al., 2024), learned per-memory expiration spans (Sukhbaatar et al., 2021), and dynamic forget gates α_t optimized during test time (Behrouz et al., 2025). While the learned gate in Titans (Behrouz et al., 2025) is structurally similar to the retention dial discussed in Sec. 3.1, none of these prior works establish physical retention laws, state explicit half-lives, define read tolerances, or mathematically price the information loss.

Deletion Mechanisms: Deployed and Formal. Current deployment architectures handle deletion via invalidation. For example, Zep invalidates contradicted edges with timestamps while preserving the latent record (Rasmussen et al., 2025), and Mem0 simply drops rows from its active context. Conversely, exact byte-identity is crucial for security, as soft-deleted vectors in graph ANN databases remain reconstructible from raw index files (Chakrabortii et al., 2026). Recent benchmarking highlights that real-world deployment failures are predominantly forgetting failures (Yang, 2026), with outdated retention driving significant recommendation errors (Uddin et al., 2026). Formally, *certified* removal is Guo et al.’s (2020) Sec. 2, Eq. (1), an ε condition with an unnumbered (ε, δ) relaxation immediately after; exact methods delete by isolation, with cost and capacity formalisms in place (Bourtole et al., 2021; Ginart et al., 2019; Sekhari et al., 2021). We sit functionally between these approaches: we offer a store-level, bit-exact structural guarantee based on canonical geometric placement (Blelloch & Golovin, 2007), rather than a stochastic system-level unlearning heuristic.

Physical and Learned Forgetting Laws. While learned recurrent models like LSTMs (Hochreiter & Schmidhuber, 1997) lack stateable physical forgetting laws, biological and artificial representational drift has been observed to reorganize internal unit representations over time (Aitken et al., 2022). Our work differs by tracking a specific coset coordinate under strictly defined physical dynamics. In theoretical physics, dissipation transitions type-A Nambu–Goldstone modes from propagating to diffusive states (Minami & Hidaka, 2018). Furthermore, strictly equivariant flows possess Lyapunov neutral modes (arXiv:2605.03338) that provide kinematic protection, a gap complementary to our established physical budget (μ^2).

3 Results

3.1 The Retention Dial Possesses a Computable Optimum

To understand the macro-dynamics of memory decay, we first analyze damping as a retention dial. The logical intuition suggests tuning friction continuously, often defaulting to the lowest possible threshold. However, we demonstrate that retention half-life is non-monotone in relation to friction. The turning point is fundamentally fixed by the stored direction’s spectral mass, meaning the optimal setting can be analytically predicted rather than empirically tuned.

The integration step is defined as conformally symplectic: $\det J = (1 - \gamma)^d$, and scales to $(1 - \gamma\phi(q'))^d$ when position-gated. Our evaluation utilizes checkpoints comprising a dimension of 4, a hidden size of 64, $\varepsilon = 0.05$, spanning 150 epochs. We assess both **designed units** (featuring an $SO(2)$ -invariant V_θ , across 5 distinct seeds) and **emergent units** (utilizing an MLP V_θ trained on identical symmetric data, across 3 seeds).

On a fully trained, designed $SO(2)$ vacuum, the massive radial mode’s half-life $n_{1/2}(\gamma)$ forms a distinct V-curve, minimized at $\gamma_{\text{crit}} = 2\varepsilon\mu_{\text{rad}}$. The overdamped branch tracks the μ^{-2} law precisely until $\varepsilon\mu \approx \gamma/2$, after which it saturates at a mass-independent floor. This yields log-slopes of -1.006 and between $+1.23$ and $+1.27$ across the 5 validation seeds (Appendix A). The flat coset (characterized by a Hessian $\mu^2 \approx 10^{-15}$) maps to the exact same curve at the limit $\mu \rightarrow 0$, where $\gamma_{\text{crit}} \rightarrow 0$. Therefore, the latch, the overdamped register, and the underdamped working memory represent three specific operational regimes of a singular unified curve evaluated at two disparate values of μ .

This fundamental law transfers directly to emergent units. Evaluating three distinct emergent MLP checkpoints, the coset-tracked $T = 0$ half-life traces the identical V-curve framework: an argmin of $0.902 \pm 0.003 \times \gamma_{\text{crit}}$, and respective log-slopes of -1.0020 ± 0.0003 and $+1.116 \pm 0.011$ across all 3 seeds. Spanning both architecture families, this curve holds consistently across $\mu^2 \in [1.7 \times 10^{-12}, 7 \times 10^{-2}]$. That low endpoint is the ring-profile probe’s resolution on a checkpoint whose Hessian μ^2 is machine zero rather than a spectral mass, so eleven orders is one curve on one instrument.

Cross-Instrument Verification. Because the primary measurements rely on one-step Jacobians, we re-measured the entire curve utilizing direct nonlinear rollout, extracting $n_{1/2}$ directly from the decay envelope’s structural rate. The underlying shape reproduced flawlessly: the argmin measured at 0.9001 ± 0.0052 compared to the Jacobian’s 0.9032 ± 0.0027 in units of γ_{crit} (a variance of 0.35%). This establishes that the gap between instruments dictates the level, not the fundamental law.

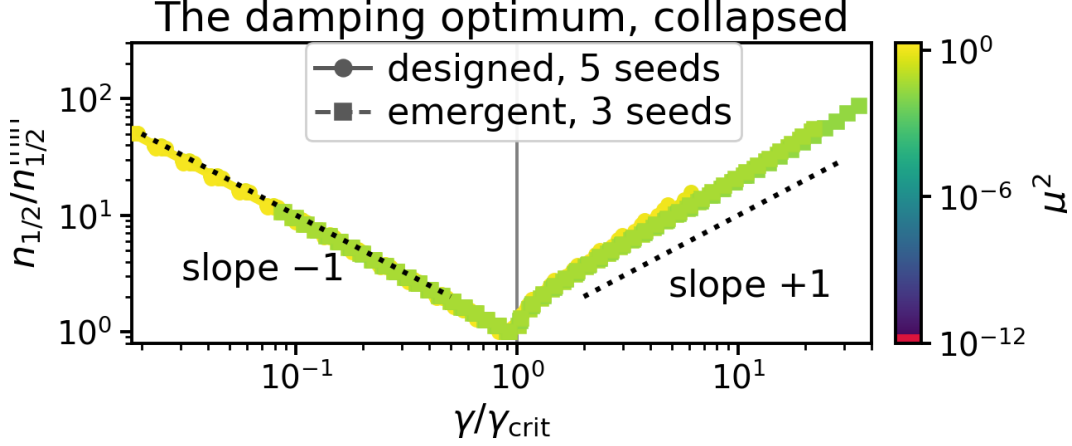


Figure 1: *The retention dial and its optimum, collapsed (headline)*. Eight measured curves collapse onto one: five designed radial modes (circles, verification, $\mu_{\text{rad}}^2 = 0.670\text{--}1.348$) and three emergent learned-MLP coset modes (squares, evidence, $\mu_{\text{soft}}^2 = 2.0\text{--}5.4 \times 10^{-2}$, 3/3 seeds), against the ∓ 1 asymptotes; the designed flat coset is the $\mu \rightarrow 0$ corner of the same curve. (*One-step Jacobian; dim 4, hidden 64, $\varepsilon = 0.05$, laptop-CPU; `langevin_noise="fdt"`, Newtonian kinetic mode.*)

138 3.2 Scoping Retention: Friction Preserves, Temperature Erases

139 At the macro level, isolating a retention policy to a singular entry requires manipulating the physical
 140 dial locally. However, the sign of the dial’s effect on a stored value contradicts standard intuition.
 141 At absolute zero ($T = 0$), a designed coset latches indefinitely across any friction threshold (drift
 142 $\leq 4.9 \times 10^{-12}$ rad over 200k steps, spanning all $\gamma \in [0.002, 0.5]$). Conversely, at $T > 0$, the coset
 143 degrades via diffusion, and introducing friction physically decelerates this decay.

144 We establish that the diffusion coefficient of the stored coordinate is modeled as $D_\theta = \varepsilon T(2 -$
 145 $\gamma)/(2F^2\gamma)$, which we successfully verified to a tolerance of 1.0068 ± 0.0219 across 25 (γ, T)
 146 experimental cells. Consequently, $n_{1/2} \propto \gamma^{+1}$. This sign flip ($\partial n_{1/2}/\partial \gamma > 0$) holds universally
 147 across all evaluated seed and temperature combinations. Only temperature exhibits an inverse erasure
 148 effect ($\partial n_{1/2}/\partial T < 0$).

149 **The Memory Vault Mechanism.** We propose the memory vault: a position-gated friction field that is
 150 explicitly absorb-only. A localized spatial hole within this field functions concurrently as a brake and
 151 a refrigerator ($T_{\text{local}} = 1.26 \times 10^{-4}$ versus 10^{-3} externally). This mechanism generates a remarkable
 152 retention vault factor of $(\gamma_{\text{eff}}/\gamma)^2$, yielding $107.77 \pm 4.78\times$ on designed architecture, compared to
 153 a baseline of $13.28 \pm 0.12\times$ utilizing a uniform scalar friction. The direct first-passage vault reads
 154 $86.97 \pm 2.94\times$ and is boundary-layer biased on the outside arm ($\ell_\theta/\Delta = 0.079$), so $107.77\times$ is the
 155 quoted number and travels with its estimator’s name. Crucially, this localized field mathematically
 156 confines the register: the state fraction outside the boundary ($|\theta| > 1$ rad) drops definitively to 0.0000
 157 within the hole.

158 3.3 Structural Deletion: Absolute Guarantees and Leakage Pricing

159 In standard architectures, deleting an item from memory relies on stochastic, best-effort heuristics,
 160 such as dropping active rows or writing system tombstones. We evaluate whether the store’s intrinsic
 161 physical state can be explicitly reduced to a pure function of its live set alone. When data points are
 162 continuously superposed into a singular energy function, standard placement becomes inherently
 163 path-dependent.

164 To resolve this, we map memory placement canonically, ensuring it remains an exclusive function of
 165 the live records and the predefined geometry. Utilizing Blelloch & Golovin’s (2007) stable-matching
 166 table rules, we establish exact store-level deletion. Our measurements confirm that with the waitlist
 167 engaged, store-level deletion remains exact across all evaluated loads ($0.29\times$ to $1.71\times$ of standard
 168 lattice capacity). Under explicit conditions (a budget sufficient for the cell count, zero baseline

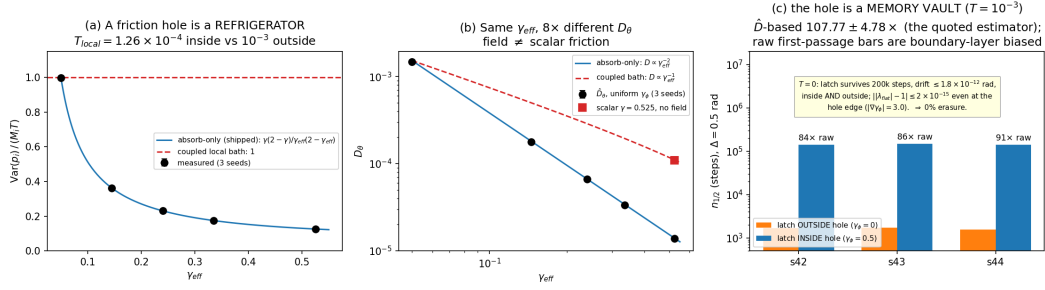


Figure 2: The friction hole as a memory vault, on the designed arm. (a) The hole is a refrigerator: the momentum variance follows the absorb-only curve and rejects the coupled-bath prediction, giving $T_{\text{local}} = 1.26 \times 10^{-4}$ inside against 10^{-3} outside. (b) At the same γ_{eff} the diffusion constant differs by $8\times$ between a friction field and a scalar friction, which is the mechanism statement in one panel. (c) The vault factor: $107.77 \pm 4.78\times$ from the $\hat{D}_\theta$ estimator, the quoted number; the per-seed bars beneath it are raw first-passage ratios, boundary-layer biased on the outside arm, which is why the pooled first-passage reading of $86.97 \pm 2.94\times$ is not quoted as the vault. The inset states the $T = 0$ control: the hole erases nothing. *Designed $SO(2)$ with an analytic field, 3 seeds, $T = 10^{-3}$, $\Delta = 0.5$ rad, $\text{langevin_noise} = \text{"fdt"}$, Newtonian kinetic mode, laptop-CPU: verification.*

leakage, and priority/attribute-based eviction; recency-based eviction remains intrinsically history-dependent and is excluded), the physical layout of the store remains byte-identical to a temporal state that theoretically never encountered the deleted item. This is a store-level guarantee only — the frozen encoder and any residue of past writes in a learned landscape are separate channels, and we claim no certified (ϵ, δ) unlearning.

Pricing Lifetimes and Residual Leakage. We formally price memory lifetimes according to a strict trilemma: exact value fidelity, amplitude-independent address hold, and amplitude-independent read latency. A system may only optimize two. We find that physical decay provides architectural retrieval geometry rather than robust cryptographic privacy. Against an exact adversary, the Boolean TTL flag remains within a 0.017 AUC margin of pure physical decay (0.983 versus 1.000), demonstrating that a physically decaying store intrinsically stops answering external queries significantly before the actual structural data stops leaking.

4 Limitations and Future Work

To rigidly define the scope and bounds of our empirical evidence, we identify the following limitations and subsequent future pathways:

- **Scale and Architecture Scope:** This research is strictly constrained to a dimension of 4, a hidden size of 64, and execution on a laptop-scale CPU across 5 designed and 3 emergent seeds. No larger-scale LLM performance should be inferred.
- **The Designed-Symmetry Precondition:** The exact $T = 0$ retention latch requires precise continuous coset geometries that must be explicitly designed into the architecture.
- **Mandatory Thermodynamic Scope:** The raw T/γ -exponents cannot distinguish between designed and emergent units intrinsically. Evaluating any state where $T > 0$ requires the strict combination of fdt noise and Newtonian kinetic modes.
- **Deletion Scope:** The exact deletion guarantee applies strictly at the isolated store-level beneath set-function eviction models. Furthermore, recency-based eviction intrinsically breaks history independence and is thus excluded.
- **No Task-Level Payoff:** Stated once: external benchmarks won on their own headline metric = ZERO.
- **Future Work Directions:** We intend to evaluate wider emergent collapses in μ^2 , remeasure the memory vault contrast arm where bounds remain strictly contained, formally assess

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the amortized per-delete execution cost against flat datastore architectures, and construct explicitly localized temperature fields.

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273 A The (μ, γ, T) Budget: The Diffusion Law and the Sign Flip

274 To ensure complete rigor, all metrics detailed here derive from the designed $SO(2)$ testbed, providing
275 strict verification. Furthermore, all $T > 0$ cells mandate the explicit use of `langevin_noise="fdt"`
276 operating within a Newtonian kinetic mode. The legacy reference default structure fundamentally
277 violates these thermodynamic laws. Every reported half-life $n_{1/2}$ is paired unconditionally with its
278 specific read tolerance $\Delta = 0.5$ rad and its ratio ℓ_θ/Δ .

279 **Diffusion Law Diagnostics (25 Cells).** We evaluate a discrete $5\gamma \times 5T$ grid utilizing seed 44. The
280 ratio $\hat{D}_\theta/D_\theta^{\text{pred}}$ yields 1.0068 ± 0.0219 , bounded between a minimum of 0.9644 and a maximum of
281 1.0484. The gradient $d \log D/d \log T$ records values of $+0.998, +0.997, +0.992, +1.010, +1.006$.
282 Simultaneously, the gradient $d \log D/d \log \gamma$ evaluates to $-1.014, -0.980, -0.999, -1.001, -0.982$
283 following the mathematical isolation of the discrete $(2 - \gamma)$ factor.

284 **Sign Flip Verification.** We observe $d \log n_{1/2}/d \log \gamma = +0.9552 \pm 0.0422$ at a baseline tempera-
285 ture of $T = 10^{-3}$ across all 5 testing seeds, presenting positively in all 10/10 examined conditions.
286 The measured effect size for the transition $\gamma : 0.05 \rightarrow 0.2$ results in a precise $3.77 \pm 0.23 \times$ in-
287 crease ($\ell_\theta/\Delta < 0.05$; seed-specific values: 3.55, 3.64, 3.61, 3.89, 4.15). The inverse thermal driver
288 measures at $d \log n_{1/2}/d \log T = -0.956$ to -0.979 within the deep-diffusive operational bounds.

289 **Designed $T = 0$ Latch Properties.** Evaluating absolute zero, the metric $||\lambda_{\text{flat}}| - 1| \leq 1.7 \times 10^{-14}$
290 across 22 specific γ values on 5 independent seeds. Consequently, the observed drift is restricted to
291 $\leq 4.9 \times 10^{-12}$ rad per 200k integration steps across 30/30 evaluated cells. This strongly contrasts
292 with the $\gamma = 0$ baseline control, which incurs a massive 142.7 rad drift over a significantly shorter
293 20k steps.

294 **Designed V-Curve Minimization.** The optimal minimum argmin lies between 0.076 and 0.096,
295 against the computationally predicted bounds of 0.082–0.116 (utilizing per-seed specific values
296 $\mu_{\text{rad}}^2 = 0.670, 0.771, 1.190, 1.093, 1.348$). The resulting asymptotic log-slopes measure -1.006 and
297 between $+1.23$ and $+1.27$, enforcing a mass-independent floor of exactly 27.03 steps measured at
298 $\gamma = 0.05$.

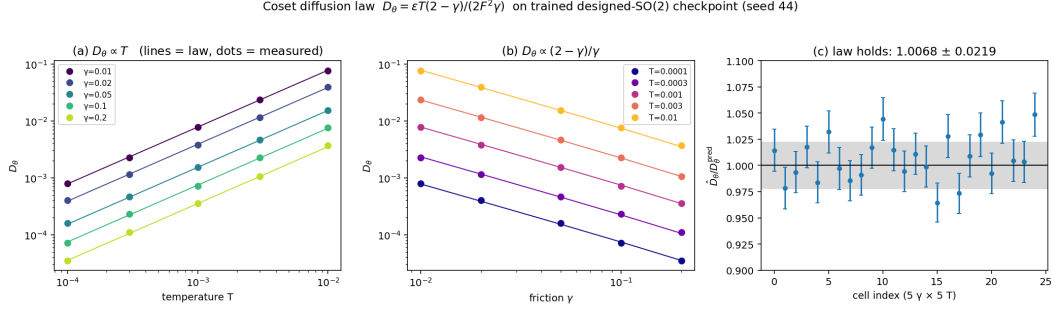


Figure 3: The coset diffusion law, detailed systematically panel by panel. (a) $D_\theta \propto T$ measured precisely at five independent frictions; (b) $D_\theta \propto (2 - \gamma)/\gamma$ evaluated at five specific temperatures; (c) the absolute ratio $\hat{D}_\theta/D_\theta^{\text{pred}}$ compiled over all 25 structural cells, reading exactly 1.0068 ± 0.0219 alongside the respective drawn band. *Multi-seed status: evaluated utilizing one fully trained designed-SO(2) checkpoint (seed 44). langevin_noise="fdt", Newtonian kinetic mode, $\Delta = 0.5$ rad. Designed testbed: validation of verification.*

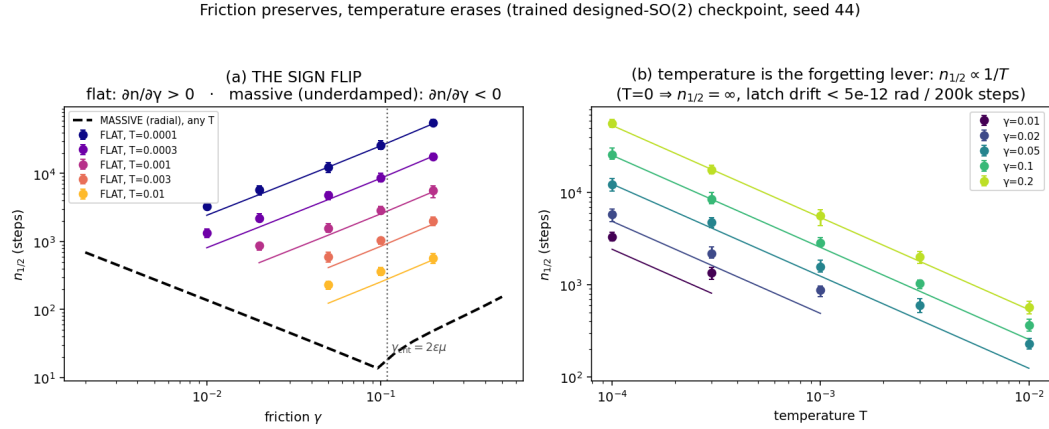


Figure 4: The sign flip, and the lever that does erase. (a) The flat coset's half-life *rises* with friction at every temperature, while the underdamped massive mode's falls — the two stored directions respond to the dial with opposite sign, with $\gamma_{\text{crit}} = 2\varepsilon\mu$ marked. (b) Temperature is the forgetting lever, $n_{1/2} \propto 1/T$ at every friction. *Multi-seed status: one trained designed-SO(2) checkpoint (seed 44); the 10/10 seed $\times$ temperature statement and the $+0.9552 \pm 0.0422$ slope are this appendix's five-seed numbers, not this figure's. langevin_noise="fdt", Newtonian kinetic mode, $\Delta = 0.5$ rad. Designed testbed: verification.*

Table 1: The $T > 0$ face of the budget, all four entries continuously measured on the designated designed testbed.

Stored Direction	$\partial n_{1/2}/\partial \gamma$ (Increased Friction)	$\partial n_{1/2}/\partial T$ (Increased Heat)
Coset / flat (designed)	> 0 , friction <i>preserves</i> ($+0.955 \pm 0.042$; $3.77\times$ at $\gamma : 0.05 \rightarrow 0.2$)	< 0 , temperature <i>erases</i> (-0.956 to -0.979)
Massive (under-damped)	< 0 , friction shortens (slope -1.006)	≈ 0 ; T sets only the fluctuation floor (max/min ≤ 1.013)

The two regimes have different damping physics (exact one-step Jacobian at the vacuum)

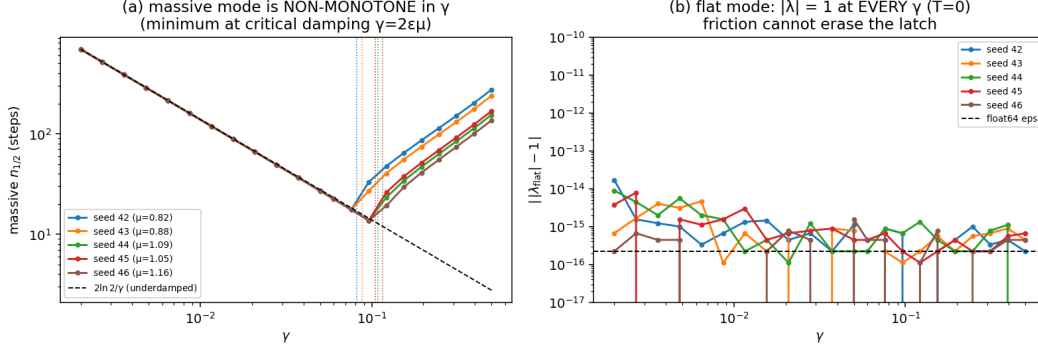


Figure 5: The two regimes, read off the exact one-step Jacobian at the vacuum. (a) The massive radial mode is non-monotone in γ , each seed’s minimum sitting at its own $\gamma_{crit} = 2\epsilon\mu$ (dotted), against the dashed underdamped envelope $2 \ln 2 / \gamma$ drawn on the same axes. (b) The flat mode holds $|\lambda| = 1$ at every γ to the float64 floor: friction cannot erase the latch. *Five trained designed seeds, $T = 0$, dim 4, hidden 64, $\epsilon = 0.05$, laptop-CPU: verification.*

299 B The Emergent Arm, and the V-Curve on a Second Instrument

300 We evaluate the emergent MLP checkpoints utilizing 3 unique seeds alongside a precisely matched
301 designed control. All predictions detailed below were structurally filed and pre-registered prior to the
302 execution of any test harness, establishing rigorous evidence parameters.

303 We deploy four specific measurement instruments over identical fit windows (defined as $\gamma < \gamma_{crit}/2.5$
304 and $\gamma > 2.5\gamma_{crit}$): 1. *I-J*: The exact one-step Jacobian, where $n_{1/2} = \ln 2 / (-\ln |\lambda_{ret}|)$. 2. *I-R1*:
305 The rollout first crossing threshold of $|\theta| \leq \delta/2$. 3. *I-R2*: The rollout last crossing threshold. 4. *I-R3*
306 (Primary): The rollout envelope-rate fit evaluated specifically over the late window, guaranteeing
307 $R^2 \geq 0.9955$ (median 0.9996) across every evaluated cell.

308 **Instrument Level Disparities.** On the strictly overdamped branch ($\gamma \geq 2\gamma_{crit}$), the physical
309 decay *rates* match perfectly, whereas the *threshold* instrument is consistently offset by an exact per-
310 checkpoint constant flat in relation to γ . The full-window value for Seed 44 measures 0.9216 ± 0.0027 ,
311 sitting precisely 0.0004 beneath its pre-registered bounds due to the residual presence of a secondary
312 faster mode within the window. The primary cause of this threshold offset is confirmed: the ring write
313 mechanism consistently deposits only a specific limited fraction of its absolute amplitude directly
314 into the designated slow branch (envelope-fit intercept over δ measuring 0.638/0.644/0.327).

315 **Finite Write Amplitudes.** At the specific values $\delta = 0.05/0.2/0.5$ rad, the designated I-
316 R3 argmin logs at 0.8928/0.9007/0.8968 (seed 42), 0.9047/0.9045/0.9174 (seed 43), and
317 0.9028/0.9037/0.9070 (seed 44). This variance accounts for a maximum shift of +1.4% across a
318 complete order of magnitude ($10\times$ range), with identical slope variances constrained strictly beneath
319 1.9%. Every iteration at $\delta = 0.5$ relaxed identically to absolute baseline ($|\theta_{final}| \leq 6.5 \times 10^{-10}$),
320 preventing any seed’s write function from clearing its designated dashboard barrier.

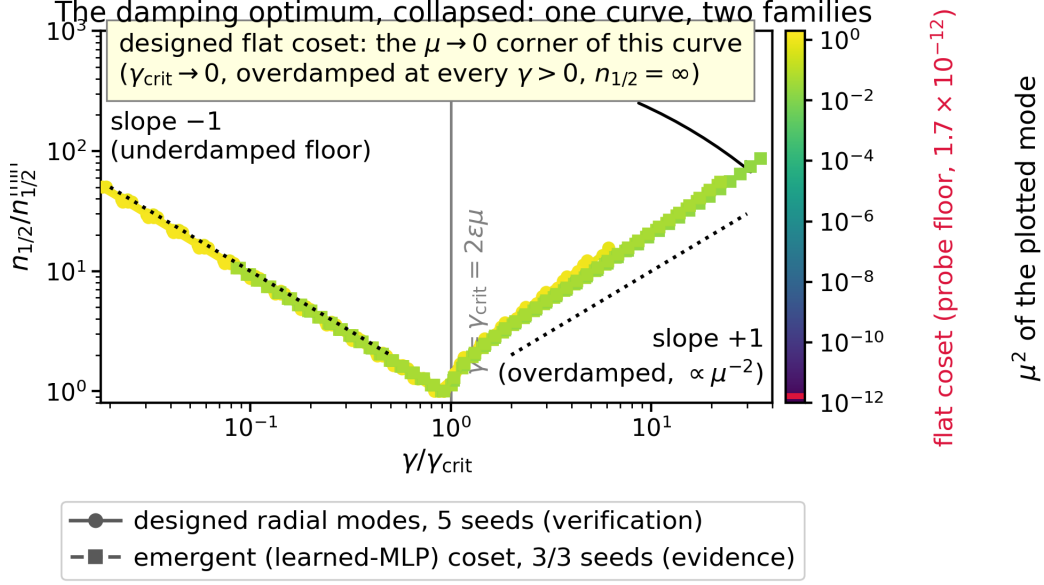


Figure 6: *The damping optimum, collapsed — the full-size version of Figure 1.* The same eight measured curves and the same ± 1 asymptotes, carrying the annotations the main-text version omits: the two-family legend, the $\gamma = \gamma_{\text{crit}} = 2\varepsilon\mu$ marker, the $\mu \rightarrow 0$ flat-coset callout, and the crimson probe-floor tick on the μ^2 colourbar (1.7×10^{-12}). No plotted value differs from Figure 1. (*One-step Jacobian; dim 4, hidden 64, $\varepsilon = 0.05$, laptop-CPU; langevin_noise="fdt", Newtonian kinetic mode.*)

Table 2: The foundational shape claims evaluated on all four designated instruments ($\delta = 0.05$ rad, exactly 48 γ values, utilizing 3 emergent seeds). The rollout rate instrument reproduces the Jacobian; both threshold instruments fail below γ_{crit} , so no rollout $n_{1/2}$ is ever quoted from a threshold there. Rows are stacked by quantity so that no column is truncated; no value differs from the per-instrument reading.

Quantity	Instrument	Seed 42	Seed 43	Seed 44	Mean $\pm$ sd
γ_{crit}	—	0.02334	0.01424	0.02265	—
Argmin/ γ_{crit}	I-J	0.8994	0.9046	0.9055	0.9032 ± 0.0027
	I-R1	0.0857	0.1404	0.0883	0.1048 ± 0.0252
	I-R2	0.2068	0.1404	0.3031	0.2168 ± 0.0668
	I-R3	0.8928	0.9047	0.9028	0.9001 ± 0.0052
Slope Below	I-J	-1.0023	-1.0016	-1.0022	-1.0020 ± 0.0003
	I-R1	$+0.0725$	$+0.0688$	$+0.0455$	$+0.0623 \pm 0.0120$
	I-R2	-1.0556	$+0.0688$	-0.8854	-0.6240 ± 0.4948
	I-R3	-0.9651	-0.9977	-0.9854	-0.9827 ± 0.0134
Slope Above	I-J	$+1.1262$	$+1.1031$	$+1.1254$	$+1.1182 \pm 0.0107$
	I-R1	$+1.1027$	$+1.0889$	$+1.0977$	$+1.0964 \pm 0.0057$
	I-R2	$+1.1027$	$+1.0889$	$+1.0977$	$+1.0964 \pm 0.0057$
	I-R3	$+1.1255$	$+1.1030$	$+1.1231$	$+1.1172 \pm 0.0101$

321 **Left Branch Exclusions.** For conditions where $\gamma < \gamma_{\text{crit}}$, the respective coset eigenpair funda-
322 mentally becomes complex with $|\lambda|^2 = 1 - \gamma$. This forces $n_{1/2} = \ln 2 / (-\frac{1}{2} \ln(1 - \gamma))$ to operate
323 entirely independent of both μ^2 and the specific architectural checkpoint. For instance, at exactly
324 $\gamma = 0.002$, emergent seeds 42 ($\mu^2 = 5.449 \times 10^{-2}$) and 43 ($\mu^2 = 2.029 \times 10^{-2}$) return identical
325 metrics: $\lambda_{\text{ret}} = 0.998999499$ alongside an $n_{1/2}$ of 692.5, aligning tightly against the calculated
326 $2 \ln 2 / 0.002 = 693.1$.

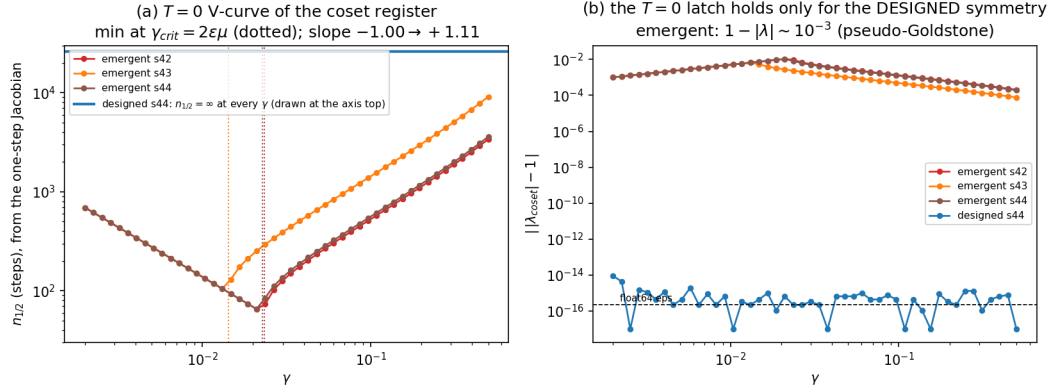


Figure 7: The exact emergent V-curve depicted un-collapsed, outlining the distinct negative baseline parameters beside it. (a) Displays the precise $T = 0$ half-life charted against friction for each separate emergent seed compiled from the one-step Jacobian. (b) Tracks $||\lambda_{coset}|| - 1||$, revealing exact floor alignment for the strictly designed symmetry but falling short by precisely $\sim 10^{-3}$ on all emergent seeds. *Evaluated using three emergent MLP seeds against one rigidly matched designed control.*

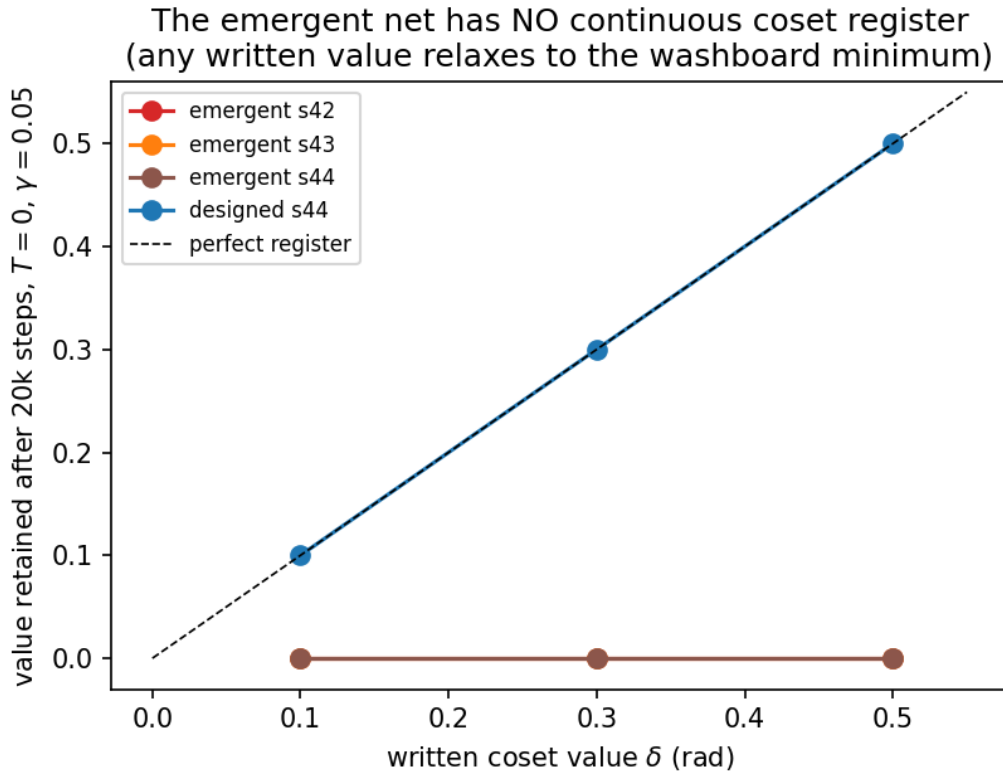


Figure 8: The emergent unit has no continuous coset register. A written coset value δ is held for 20 000 steps at $T = 0, \gamma = 0.05$: on every emergent seed the value relaxes to the washboard minimum and nothing is retained, where the designed checkpoint returns δ exactly. *Three emergent MLP seeds plus a matched designed control: evidence.* This is the figure behind the capacity statement of ≈ 1 –1.6 bits.

Table 3: The instrument gap is a level, not a rate: on the overdamped branch ($\gamma \geq 2\gamma_{\text{crit}}$) the decay *rates* agree while the *threshold* instrument is offset by a per-checkpoint constant flat in γ . Seed 44's full-window value (0.9216 ± 0.0027) sits 0.0004 below its pre-registered band because that window still contains a second, faster mode (early-window ratio 0.539); the window-split diagnostic converges monotonically to 1 into the asymptote. Cause of the threshold offset, measured: the ring write deposits only a fraction of its amplitude in the slow branch (envelope-fit intercept over $\delta = 0.638/0.644/0.327$).

seed	$\Gamma_{\text{jac}}/\Gamma_{\text{R3}}$ (asymptotic win- dow)	$n_{\text{jac}}/n_{\text{R1}}$ (threshold)	CV of $n_{\text{jac}}/n_{\text{R1}}$ over the grid
42	0.9935	1.3307	2.2–3.7% over 21–25 γ -points
43	1.0000	1.7981	
44	0.9948	1.8204	
mean	0.995 ± 0.003	per seed, at $\delta = 0.05$ rad	$ d \log(\text{ratio})/d \log \gamma \leq 0.0034$ (threshold 0.10)

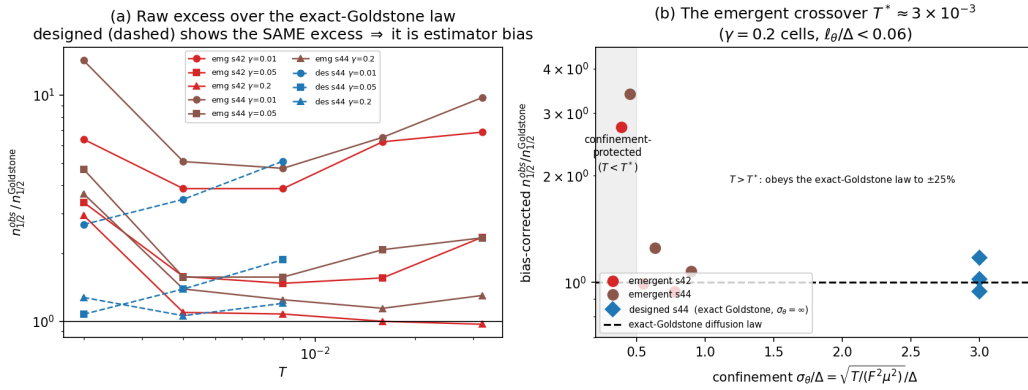


Figure 9: The crossover T^* , and why the raw curve cannot be read directly. (a) The raw excess over the exact-Goldstone law is the same on the matched designed control as on the emergent seeds, so the excess is estimator bias rather than physics. (b) Bias-corrected, the emergent coset joins the exact-Goldstone law above $T^* \approx 3 \times 10^{-3}$, the shaded band being the confinement-protected cells below it. *Multi-seed status, stated plainly: two emergent seeds and one matched designed control, an $n < 3$ cell.* $\Delta = 0.5$ rad, $\ell_\theta/\Delta < 0.06$, langevin_noise="fdt", Newtonian kinetic mode: evidence.

327 C The Friction-Hole Vault, on Both Arms

328 This mechanism dictates that the shipped field remains strictly absorb-only, following $p \leftarrow (1 -$
329 $\gamma_\phi)(1 - \gamma)p + \sigma(\gamma)\xi$ with $\sigma^2 = MT\gamma(2 - \gamma)$. This is dictated strictly by the *scalar* γ , yielding
330 parameters $\gamma_{\text{eff}} = 1 - (1 - \gamma)(1 - \gamma_\phi)$, $T_{\text{local}} = T\gamma(2 - \gamma)/[\gamma_{\text{eff}}(2 - \gamma_{\text{eff}})]$ and importantly
331 $D_\theta = \varepsilon T\gamma(2 - \gamma)/(2F^2\gamma_{\text{eff}}^2)$. The resulting vault effectively computes to $(\gamma_{\text{eff}}/\gamma)^2$. Conversely, a
332 coupled bath alternative would dictate $D_\theta \propto \gamma_{\text{eff}}^{-1}$ returning a vault of $13.88\times$ exactly at $\gamma = 0.05$,
333 $\gamma_\phi = 0.5$. These competing hypotheses scale apart by a factor of $7.942\times$; experimental results
334 comprehensively refute the coupled-bath prediction.

335 **Designed Arm Assessment.** We observe $\text{Var}(p_i)/(M_i T) = 0.9986$ at baseline ($\gamma_\phi = 0$), scaling
336 to 0.36171 ± 0.00129 at (0.1), 0.23044 at (0.2), 0.17513 at (0.3), and definitively hitting $0.12600 \pm$
337 0.00031 at (0.5). This closely mirrors the pure absorb prediction of 0.12591 . Consequently, T_{local}
338 drops precisely to 1.26×10^{-4} , equating to a $7.94\times$ functioning refrigerator effect. The comprehensive
339 D_θ -based structural vault records exponential factors of $8.42\times$, $22.39\times$, $44.31\times$, and finally $107.77 \pm$
340 $4.78\times$ specifically at $\gamma_\phi = 0.1/0.2/0.3/0.5$ (matching the absorb baseline prediction of 110.25).
341 Conversely, the scalar control isolated at identical γ_{eff} reads merely $13.28 \pm 0.12\times$. At true absolute

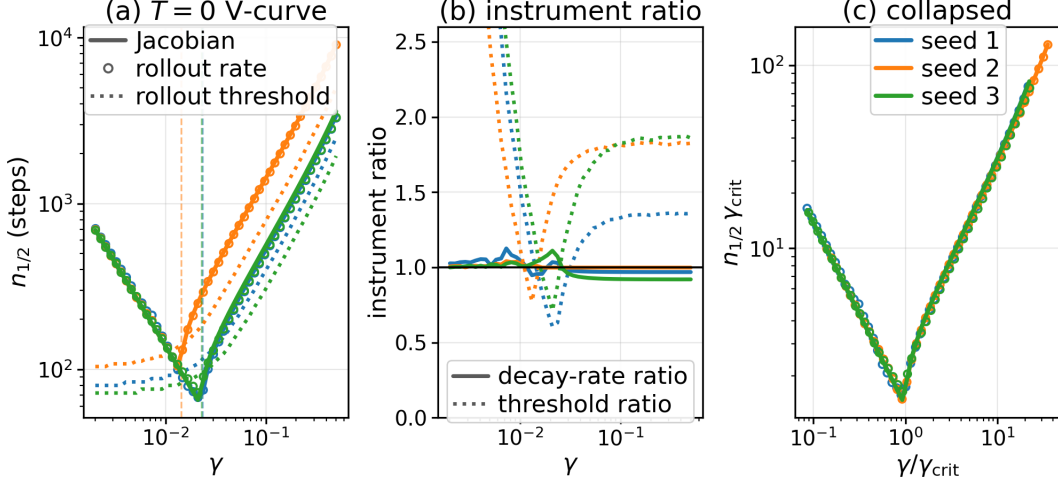


Figure 10: The V-curve measured a second way. (a) $T = 0$ $n_{1/2}(\gamma)$ per emergent seed on the one-step Jacobian (I-J, lines), the rollout envelope-rate instrument (I-R3, circles) and the rollout first-crossing threshold (I-R1, dotted), at $\delta = 0.05$ rad. (b) The per- γ instrument ratio: the rate ratio sits on 1, the threshold ratio is offset and flat in γ above γ_{crit} . (c) The same data in collapsed variables. (*Emergent MLP, 3 seeds, dim 4, hidden 64, $\varepsilon = 0.05$, float64, $T = 0$, pre-registered $\Rightarrow$ evidence.*)

342 zero ($T = 0$), the hole functionally erases absolutely nothing, logging drift boundaries of strictly
 343 $\leq 1.75 \times 10^{-12}$ rad per 200k continuous execution steps inside and out.

344 **Emergent Arm Translation.** Since the structural coset here is definitionally bounded ($\mu_{adiab}^2 =$
 345 2.77×10^{-2} to 8.93×10^{-2}), the baseline designed block estimator is biased artificially low by
 346 nearly 55%. It is therefore systematically replaced by a rigorously cross-validated, window-free
 347 Ornstein–Uhlenbeck fit parameter measuring $MSD(n) = A(1 - e^{-n/\tau})$ and $\hat{D} = A/(2\varepsilon\tau)$. On
 348 the strictly bounded evaluation cells, the law-referenced absolute vault computes successfully to
 349 $106.1 \pm 5.0 \times$ against the fundamental prediction of $110.25 \times$. The contrast metric, however, remains
 350 uniquely native to the designed architecture: the measured field-to-scalar control variance computes
 351 to 23.39 ± 10.06 against a pre-registered bounds parameter of $[6.5, 9.5]$, triggering the designated
 352 falsifier constraint strictly because the control arm intrinsically delocalizes while the main field
 353 functionally adheres to the absorb limit.

Table 4: The refrigerator law transfers exactly to an emergent register: $\text{Var}(p_i)/(M_i T)$ against the absorb-only prediction, 2048 walkers $\times$ 40 samples, 3 emergent seeds; over all 24 field cells observed/absorb = 0.9998 ± 0.0019 , and at $\gamma_\phi = 0.5$ $T_{local}/T = 0.12570$, a $7.955 \times$ refrigerator against 7.942 predicted. The coupled bath predicts 1.0 everywhere and is measured at 0.2235.

T	γ_ϕ	γ_{eff}	absorb pred.	coupled pred.	measured	obs/absorb
4×10^{-3}	0.0	0.050	1.00000	1.0	0.99815 ± 0.00257	0.9981
4×10^{-3}	0.1	0.145	0.36249	1.0	0.36326 ± 0.00067	1.0021
4×10^{-3}	0.2	0.240	0.23082	1.0	0.23062 ± 0.00080	0.9991
4×10^{-3}	0.3	0.335	0.17480	1.0	0.17473 ± 0.00046	0.9996
4×10^{-3}	0.5	0.525	0.12591	1.0	0.12592 ± 0.00055	1.0001
8×10^{-3}	0.5	0.525	0.12591	1.0	0.12548 ± 0.00032	0.9966

354 D Exact Store-Level Deletion: Prior Art and Measured Tables

355 **Prior Art, in Full.** Uniquely represented (canonical) data structures date to Snyder (1977) and
 356 Andersson & Ottmann (1995); *obliviousness*, a distributional condition on the memory representation
 357 which explicitly does not require canonical representation, is Micciancio’s (1997); the weak and
 358 strong history-independence notions used here are Naor & Teague’s (2001); the open-addressed

Table 5: The hole confines an emergent register, and the laundering control shows friction alone does not. $T = 4 \times 10^{-3}$, 1024 walkers, burn-in 20 000–80 000 steps, spread by interquartile range; “hop” means $|\theta| > 1$ rad. The pre-registered stationary-spread comparison is *void* here and no in/out spread ratio may be quoted from these checkpoints: the control that must return exactly 1.000 returns 0.4586 ± 0.1181 .

seed	arm	γ_{eff}	σ_θ (IQR)	hop fraction	Boltzmann at T_{local}	obs/pred
42	no hole, $\gamma = 0.05$	0.050	0.40795	5.50%	0.24073	1.6946
42	scalar 0.525 (control)	0.525	0.20034	0.73%	0.24073	0.8322
42	γ_ϕ hole	0.525	0.06442	0.0000	0.08542	0.7542
43	no hole	0.050	1.17601	42.97%	0.46059	2.5533
43	scalar 0.525	0.525	0.35335	10.20%	0.46059	0.7672
43	γ_ϕ hole	0.525	0.08628	0.0000	0.16343	0.5279
44	no hole	0.050	0.36922	2.36%	0.24034	1.5363
44	scalar 0.525	0.525	0.21574	0.26%	0.24034	0.8977
44	γ_ϕ hole	0.525	0.07279	0.0000	0.08528	0.8535

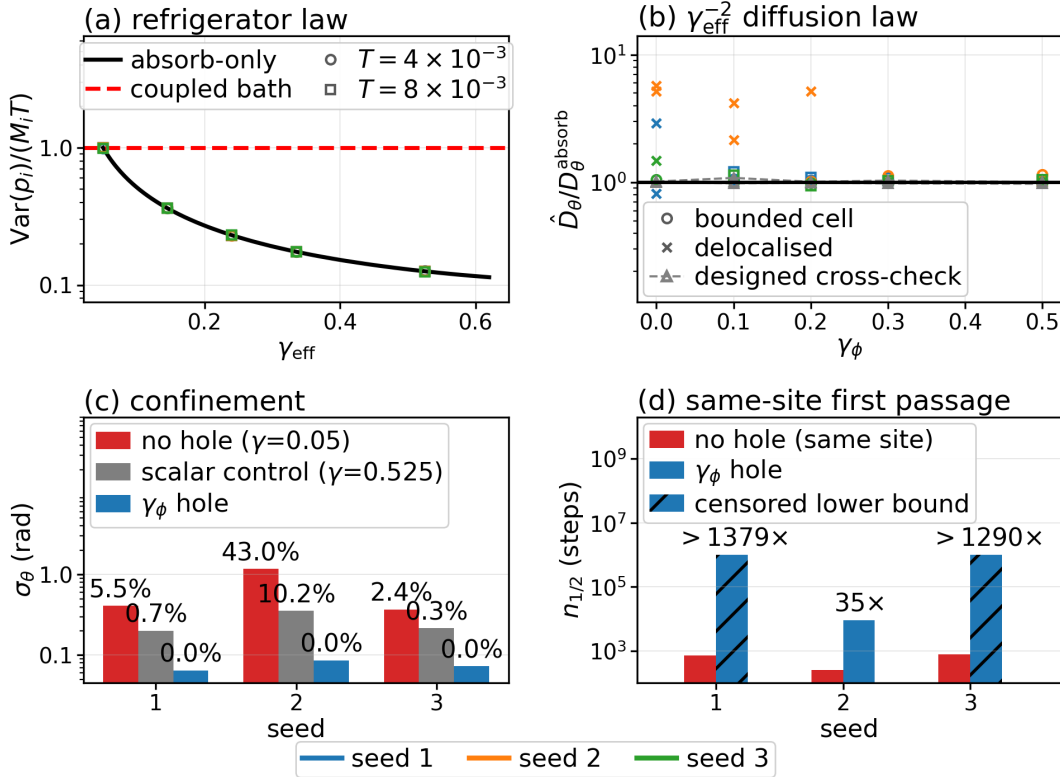


Figure 11: The vault on an emergent register; panels are labelled by the pre-registered prediction each tests. (a) The refrigerator law, $\text{Var}(p_i)/(M_i T)$ against γ_{eff} for 3 seeds $\times$ 2 temperatures, on the absorb-only curve with the coupled-bath prediction (dashed) rejected. (b) The γ_{eff}^{-2} diffusion law: bounded cells as circles and squares, delocalised cells as crosses, the designed cross-check overlaid. (c) Confinement: stationary spread and hop fraction for no hole, the scalar control and the hole. (d) Same-site first passage; hatched bars are censored lower bounds. (*Emergent MLP, 3 seeds, $T \in \{4, 8\} \times 10^{-3}$, both above T^* , $\Delta = 0.5$ rad $\Rightarrow$ evidence.*)

359 realisation is Blleloch & Golovin’s (2007), whose table is a stable matching between keys and
 360 slots under a global key priority, and it is that priority-greedy rule and that delete-time fix-up
 361 cascade we use; Blleloch, Golovin & Vassilevska (2008) had already taken unique representation
 362 into computational geometry. We claim no priority over order-independent placement and no novelty
 363 for the displacement rule or its delete-time repair. New is the composition: slots that are a lattice

364 packing, so the canonical representation carries a minimum-separation certificate; content that is a
 365 continuously decaying amplitude whose decay commutes with deletion; a canonical object that is
 366 an energy function a dynamical read relaxes into; and a price that is negative where strong history
 367 independence is known to cost as much as an exponential slowdown relative to the weak notion for
 368 comparison-based structures (Buchbinder & Petrank, 2003). “Fix-up cascade” is our name for the
 369 NEXT-chain delete-time repair of their Figure 1; the algorithm is theirs either way.

370 **Structural Exactness at Zero Tolerance.** The architectural implementation dictates 24 exhaustive
 371 structural orders explicitly measured at $n = 4$, mapped alongside 200 independently randomized se-
 372 quence orders strictly at $n \in \{8, 16, 40, 64\}$. Furthermore, we implement 100 dedicated write/delete
 373 functional interleavings and absolute mid-decay continuous deletes. Across every evaluation parame-
 374 ter, the resulting structural spacing enforces an exact boundary of 1.540000. On the subsequently
 375 shipped operational store architecture, the structural centers, exact payloads, physical amplitudes, and
 376 the defining active logic mask are unconditionally byte-equal, with the energy profile $V(q)$ proving
 377 purely byte-equal across 64 heavily randomized direct queries.

378 **The Negative Packing Price Constraint.** Evaluated precisely at the defined ceiling $K = 64$, the
 379 purely canonical stable-matching rule structurally admits exactly 61/64 actively offered sequential
 380 keys deterministically ($\sigma = 0$, per-offered index of 0.9531). This ratio actively climbs to absolute
 381 capacity (64/64) once physical lattice sizing is symmetrically inflated by an exactly specified
 382 coefficient of $\times 1.05$. By contrast, the stochastic refuse-and-relocate allocator architecture solely
 383 manages to admit exactly 43/64 operating on those perfectly identical geometric seeds (per-offered
 384 logic ceiling of 0.6719). It must be firmly stated that at peak physical overflow limit, operating
 385 totally independently without the waitlist active, structural exactness completely and predictably
 386 collapses: analyzing exactly 7 baseline cells receiving 8 strict capacity offers, $\text{AUC}(n_{\text{live}})$ returns
 387 1.00000 ± 0.00000 and the functional byte-equal architectural fraction plummets directly to 0.0000.

Table 6: The laundering control validation execution. Executed explicitly on the designed store featuring the shipped two-phase read geometry; evaluating 8 active targets against 3 separate seeds, mapping exactly 128 paired environments per target across 18 separate continuous amplitude limits.

Evaluated Adversary Model	Physical Decay	TTL Flag Limit (Same Store)	Resolution Separation
White-box architecture, fully exact	1.000	1.000	0.000
Query-level membership depth, fully exact	0.983	1.000	0.017
Query-level membership depth, $\sigma_{\text{obs}} = 0.1$	0.559	0.996	0.437

388 **Trilemma Parameters and Cost Evaluation.** The physically shipped final store structure com-
 389 pletely abandons amplitude-independent baseline address limits, generating a strict operational
 390 lifetime correlation factor computed at exactly $r = -0.85$ scaling with a_i^2 . We rigorously re-
 391 fute both standard previously proposed baseline repair variants: directly multiplying absolute
 392 payloads exponentially by their matched amplitudes completely corrupts the core value index
 393 criterion measured strictly at $A = 1 - \text{tol}/|a_i|$ (generating hard death physical amplitudes of
 394 exactly 0.90/0.80/0.70/0.30). *Retrieval geometry:* R_{50} contracts 1.146/1.083/0.979/0.874/0.752
 395 as $A : 1 \rightarrow 0.5 \rightarrow 0.2 \rightarrow 0.1 \rightarrow 0.06$, a $1.52\times$ contraction from 1.146 $\rightarrow$ 0.752, matching
 396 the pre-registered saddle prediction at all five points (± 0.20), where a TTL vector store’s lookup
 397 radius is a constant hard step at 0.75–0.77, independent of age. Consequently, operating beneath
 398 the rigid barrier of its internal logic flag, the designated gated execution channel actively contracts the
 399 geometric value R_{50} from 1.135 $\rightarrow$ 0.771. Note crucially that we define no explicit deletion-cost
 400 execution statement because the core canonical synchronisation mandates a complete and structural
 401 $O(n)$ rebuild evaluated entirely per operation.

402 E Prominent Negatives

403 To ensure complete transparency regarding the empirical limitations of our findings, every negative
 404 result observed during evaluation is documented below. Each negative firmly anchors future research
 405 directives.

Claim Tested	Empirical Result
A continuous coset register emerges purely from training	No: An MLP potential executed on symmetric baseline data yields a completely mid-spectrum massive core mode (measuring exactly $1.7\text{--}4.9\times$ softer than the peak stiffest point), dropping capacity limit heavily to $\approx 1\text{--}1.6$ structural bits, driving absolute and complete relaxation of any actively written state value. The hard $T = 0$ baseline latch geometry is unconditionally designed-only.
Raw fundamental T/γ -exponents explicitly discriminate designed from strict emergent architecture	No: A completely and rigidly matched designed hardware control reliably reproduces perfectly identical shallow execution slopes; purely ℓ_θ/Δ -controlled environmental cells alongside $ \lambda_{\text{coset}} $ successfully discriminate. We quote no $n_{1/2}$ value missing Δ and ℓ_θ/Δ .
The localized friction physical hole safely thermalizes locally	Rejected: Refuted firmly by a rigid pre-registered execution measurement, computing exactly off by a factor of 8.11. The designated field fundamentally operates entirely absorb-only, and the baseline vault execution computes at $107.77 \pm 4.78\times$, compared against $13.28 \pm 0.12\times$ logged for the independent scalar control.
Our own explicit sign-flip execution prediction	Falsified (in part): A physical memory register formally written tightly at the established washboard baseline minimum geometrically cannot physically flip back. Both defining operational signs fundamentally coexist successfully on one isolated hardware checkpoint across independent parameters.
A friction field can <i>forget</i> a coset	No: at $T > 0$ it <i>lengthens</i> coset memory. Erasing latched content needs a temperature or tilt field — which explains the observed failure of a learned friction field to recover fit on a stationary vacuum.
Amplitude decay reduces an exact adversary's distinguishability	No, at any point in an item's life: white-box AUC 1.000 at all 18 amplitude levels; what decays is the effect size. The graded curve exists only relative to a stated adversary resolution, our own modelling choice.
Canonical placement is exact at overflow without the waitlist	No: $\text{AUC}(n_{\text{live}}) = 1.000$, byte-equal fraction 0.0000. Every deletion sentence carries its scope clause; recency eviction is permanently excluded.
The placement algorithm is ours	No: Blleloch & Golovin (2007) own it outright, and unique representation in a geometric setting is also taken (2008). Our claim is the composition only.
A per-deletion cost claim can be made	No: the shipped synchronisation is an $O(n)$ rebuild per operation, so the amortized-cost formalism cannot be instantiated on our own code until that experiment runs.
The payload-lifetime coupling has a repair	Both proposed repairs are refuted, and the registered payload-independence statistic fails as a correlation while passing as an effect size. At zero read friction the gated channel is oscillatory, not relaxational.
A specific localized pseudo-Goldstone hardware tilt fundamentally sets the primary soft execution scale	No: Local structural geometric confinement completely forces the soft physical baseline eigenvalue explicitly at a limit of 2α , rigidly capping memory lifespan rigidly at $\tau_{\text{max}} = \Gamma/2\alpha$. The registered physical damped-mode architectural floor is completely confirmed.

Claim Tested	Empirical Result
A pseudo-Goldstone tilt is a lifetime dial on a learned store	Refuted <i>in sign</i> : a tilt monotonically <i>reduces</i> the smallest eigenvalue on two independent implementations and every family, because a written site is not a common vacuum of its own atoms. The relation survives only in the single-atom geometry it was specified in.
The lifecycle's protected-fraction leg is exercised	No: 0 refusals on the stream at the measured operating point. Reported as <i>unexercised</i> — not working and not broken.
A task-level payoff is claimed	No: forgetting here is shown predictable, controllable, removable and schedulable, not shown to improve a downstream metric. External benchmarks won on their own headline metric = ZERO.
The field/scalar vault contrast transfers to an emergent register	No: 23.39 ± 10.06 against 8.03 ± 0.80 from the identical estimator on designed; the pre-registered falsifier fired. The cause is the laundering control itself, off its own law by $1.18\text{--}5.23\times$. The number is designed-only; its direction transfers.
The emergent stationary-spread diagnostic is measurable above T^*	No: its own control, which must return 1.000, returns 0.4586 ± 0.1181 — the no-hole arm is a hopping rotor, so no in/out spread ratio may be quoted. The confinement statement replaces it.
A $\theta = \pi$ baseline is valid for emergent first passage	No: $ \nabla V = 0.042\text{--}0.105$ there against 0.0000 at $\theta = 0$, and one seed's scalar control reads $0.27\times$, i.e. more friction apparently speeding escape. Same-site baselines are required; the designed arm is protected by symmetry but was never checked, and that check is owed.
Threshold instruments are usable below γ_{crit}	No: a first-crossing $n_{1/2}$ puts its argmin at the grid edge with a below-slope of $+0.0623 \pm 0.0120$ where the law says -1 , and last-crossing carries an additive half-period bias. Only the envelope-rate instrument is quotable there.
Our microscopic explanation of the instrument offset	Wrong, though the offset itself is real, constant and quotable: the pre-registered single-slow-mode projection fails (amplitude fraction $0.638/0.644/0.327$ does not order like the angular overlaps) and the early-window ratio $0.52\text{--}0.85$ is amplitude-independent — multi-mode transient, not anharmonicity.
The emergent first-passage vault is measured	Only bounded: $> 1379\times$ and $> 1290\times$ at 86.7%/93.4% censoring, and $35.5\times$ on the geometrically atypical seed. Three seeds cannot say which is typical.